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Human Values and Ethics

Report

THE FINE PRINT OF NUTRITION:

"A Deep Dive into Label Literacy, Public Health, and the Case for Stricter Food Regulation in India"

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Abstract

India is currently traversing a perilous "Nutritional Transition," where traditional diets are rapidly being replaced by ultra-processed foods (UPFs), contributing to a surge in non-communicable diseases (NCDs) like Type 2 diabetes and obesity. This report investigates the systemic "Information Asymmetry" between corporate marketing strategies and nutritional reality in the Indian FMCG sector. Through a dual-research approach, comprising an extensive literature review of FSSAI 2020 regulations and a primary survey of 51 urban respondents, we quantify the "Label Literacy" gap. Key findings (refer to Appendix A) reveal that while around 82% of consumers rely on front-of-pack claims or brand legacy, fewer than 24% understand technical terms in ingredient lists (such as Maltodextrin, INS 150d, High fructose corn syrup). Case studies on hidden sugars (maltodextrin), grain deceptions (caramel-coloured brown bread), and global quality double standards (Nestle Cerelac and spice contamination) highlight how brands exploit technical jargon and regulatory loopholes. We argue that current labelling systems, such as the proposed Indian Nutrition Rating (INR), are insufficient compared to "Warning Labels" used internationally. The report concludes with policy proposals for "Plain Language Labelling" and a layered "Sugar Tax," ultimately framing nutritional transparency not just as a regulatory issue, but as a fundamental human right and ethical imperative.

Keywords— Information Asymmetry, Label Literacy, FSSAI 2020 Regulations, Ultra-Processed Foods (UPF), Nutritional Transition, Front-of-Pack Labelling (FOPL).

1. Introduction

1.1 Background of the Study

The Indian dietary ecosystem is currently undergoing a profound "Nutritional Transition," a systemic shift where indigenous, whole-food-based diets are being marginalised by the rapid proliferation of “ultra-processed foods (UPFs)”. This evolution is closely tied to the nation's economic expansion and urban growth; however, the convenience of modern packaged goods comes at a staggering metabolic cost. India is now widely recognised as the "Diabetic Capital of the World," with over 10 crore citizens diagnosed with the condition and a massive population in the pre-diabetic stage.

Our analysis identifies a significant structural failure at the centre of this crisis: the widening chasm between corporate marketing narratives and the underlying nutritional facts. While the “Food Safety and Standards Authority of India (FSSAI)” has updated its regulatory framework for disclosures, the technical "fine print" remains largely undecipherable to the public. This research investigates the current state of "Label Literacy" among Indian consumers and the ethical ramifications of deceptive branding practices within the FMCG sector.

1.2 Problem Statement

The primary obstacle identified in this study is the pervasive "Asymmetry of Information" within the food market. Most consumers base their purchasing decisions on persuasive Front-of-Pack (FOP) claims such as "100% Atta," "No Added Sugar," or "Natural Flavours, “0% Trans Fat”, without the requisite knowledge to verify these assertions via the ingredient

list (see Appendix A). Preliminary findings (see Appendix D) suggest that:

1. **Technical Obfuscation:** Manufacturers utilise sophisticated jargon (e.g., Maltodextrin, INS 150d) to mask the presence of high-glycaemic or synthetic components.
2. **Nutritional Double Standards:** A significant "Nutritional Divide" exists where global brands provide higher-quality formulations in Western markets while offering inferior, higher-sugar, or cheaper-oil variants in India.
3. **Regulatory Complexity:** The current FSSAI labelling standards are overly complex, facilitating the creation of "Health Halos" that mislead even highly educated demographic segments.

1.3 Objectives of the Research

The core objectives of this investigation are:

- **To Quantify Label Literacy:** Utilising primary survey data from 51 respondents to determine the percentage of the public that accurately interprets "Hidden Sugars" and "INS Codes".
- **To Analyse the 'Trust-Reality' Gap:** Pinpointing which product categories (e.g., Biscuits, Juices, Breads) suffer from the highest levels of misplaced consumer trust based on misleading FOP claims.
- **To Evaluate the Case for Reform:** Assessing whether Indian consumers would prefer and better utilise "Traffic Light" warning symbols or "Stop Sign" labels over the existing complex numerical tables.

- **To Highlight Corporate Ethics:** Examining the moral obligation of multi-national brands to maintain consistent ingredient quality across global borders.

1.4 Scope and Methodology Overview

This report employs a dual-research methodology. First, it presents a comprehensive Literature Review covering the FSSAI 2020 regulations and some critical case studies on industry deception. Second, it synthesises primary data from a survey conducted by our team that collected 51 detailed responses from a diverse cross-section of students, parents, and working professionals. This evidence-based approach moves beyond theoretical critiques to provide a data-driven "Deep Dive" into the reality of public health awareness and regulatory efficacy in India.

2. Detailed Literature Review & Case Studies

2.1 Unpacking FSSAI 2020 and the Weight Principle

“The Food Safety and Standards (Labelling and Display) Regulations”, 2020, aimed to harmonise Indian standards with international benchmarks like the Codex Alimentarius. However, technical nuances often remain hidden. A critical requirement under Regulation 5(2)(b) is that ingredients must be listed in descending order of their composition by weight or volume at the time of manufacture. If "Refined Wheat Flour" or "Sugar" appears first, it is the dominant component. Manufacturers often circumvent this by splitting "Sugar" into multiple technical names (e.g., maltodextrin, invert sugar, high

fructose corn syrup) so that no single "sugar" appears as the first ingredient, despite the total glycaemic load being massive.

2.2 Decoding INS Codes and Class Titles

FSSAI allows the use of "class titles" and International Numbering System (INS) codes, which frequently obscure the true nature of additives. For example, "Edible Vegetable Oil" can represent anything from sunflower oil to cheap, high-saturated-fat palm oil. Similarly, codes like "INS 150d" (Caramel Colour IV) or "INS 621" (Monosodium Glutamate) are viewed as laboratory jargon by consumers, despite ongoing debates regarding their impact on metabolic health.

To assist readers in decoding the "Fine Print," Table 1 lists the most common synthetic additives found in Indian UPFs and their associated health risks.

Table 1: Technical Analysis of Common Food Additives (INS Codes)

INS CODE	NAME	FUNCTION	DOCUMENTED HEALTH RISKS
INS 150D	Caramel Colour IV	Colorant	Ammonia-process; contains 4-MEI, a potential carcinogen
INS 621	Monosodium Glutamate	Flavour Enhancer	"Chinese Restaurant Syndrome", triggers headaches/nausea.
INS 211	Sodium Benzoate	Preservative	Forms benzene (carcinogen) with Vit C; linked to ADHD.
INS 102	Tartrazine	Yellow Dye	Linked to asthma and hyperactivity in children.
INS 330	Citric Acid	Acidity Regulator	Often synthetic (mould fermentation); can erode tooth enamel.
INS 320	BHA	Antioxidant	Classified as "possibly carcinogenic" to humans.

2.3 The 'Sugar-Free' Deception

The "Sugar-Free" or "No Added Sugar" claim is perhaps the most dangerous marketing tactic for India’s diabetic population. Consumers frequently believe these products are safer than those containing table sugar (see Appendix A), when in reality, they often contain hidden carbohydrates with a higher glycaemic index (GI).

2.3.1 The Maltodextrin Mystery

Many health drinks and "diabetic-friendly" biscuits use Maltodextrin to maintain texture and sweetness. While technically a complex carbohydrate and thus not listed as "Sugar" on the label, its metabolic impact is more severe than table sugar. Maltodextrin has a GI of 85 to 110, whereas sucrose (table sugar) has a GI of 65. This causes a more rapid spike in blood glucose, triggering an aggressive insulin response that promotes fat storage and metabolic syndrome.

Table 2: Glycaemic Indices of Table Sugar vs. Hidden Sugars

CHEMICAL / INGREDIENT NAME	COMMON LABELLING JARGON	GLYCEMIC INDEX (GI)	HEALTH RISK LEVEL
MALTODEXTRIN	Thickener, Bulking Agent	105 - 110	Very High (Higher than Glucose)
GLUCOSE (DEXTROSE)	Corn Sugar	100	Very High (Reference standard)
HIGH FRUCTOSE CORN SYRUP	HFCS, Isoglucose	87	High (Linked to fatty liver)
SUCROSE	Table Sugar	65	Moderate (The standard "Sugar")
SORBITOL	Polyol, Sugar Alcohol	4 - 9	Low (Can cause gastric distress)
STEVIA	Natural Sweetener	< 1	Negligible (Safe alternative)

2.4. The '100% Atta' Illusion and Grain Deception

2.4.1 The Brown Bread Scam

Many "Brown Breads" sold in India achieve their dark appearance not through whole grains, but through the addition of Caramel Colour (INS 150). This is an aesthetic adulteration designed to make refined flour (Maida) look like whole wheat (Atta).

2.4.2 Maida vs. Atta Percentages

Until the 2023 FSSAI amendment, a product could be marketed as "Whole Wheat" with as little as 10-15% actual whole wheat flour. New norms now require at least 75% whole wheat for the "Whole Wheat" label and 50% for "Brown Bread". However, "Brown Bread" or "Wheat Bread" only requires a 50% threshold. This means a consumer buying "Wheat Bread" for health reasons is often consuming a product that is half Maida. Our survey (see Appendix A) revealed that 62.2% of respondents trust Brown bread for their family's health, as the colour makes them believe it's better than white bread.

2.5. Global Double Standards & Contamination

2.5.1 The Nestle Cerelac Discrepancy

A 2024 investigation revealed that Nestle's Cerelac baby food in India contains an average of 2.7 grams of added sugar per serving, while the same brand sold in the UK and Germany contains zero added sugar [10] [12]. This "Nutritional Downgrade" for developing nations is an ethical crisis, as early sugar exposure establishes lifelong preferences that drive the obesity epidemic.

2.5.2 Spice Contamination: Masala vs. Export Standards

In 2024, brands like MDH and Everest faced bans in Hong Kong and Singapore due to excessive levels of Ethylene Oxide (ETO), a carcinogenic pesticide. While ETO is used as a fumigant to prevent microbial growth, residues are strictly limited internationally (0.02-0.1 mg/kg in the EU). Alarmingly, FSSAI recently increased the domestic limit for pesticide residues in spices to 0.1 mg/kg—a ten-fold increase. Furthermore, industrial dyes like Sudan I and IV, which are carcinogenic, are still frequently detected in unbranded Indian spices.[13] [14]

2.6 Dairy Deception and Vegetable Oil Substitution

2.6.1 Ice Cream vs. Frozen Dessert

Under FSSAI Regulation 2.1.7, "Ice Cream" must be made exclusively from milk fat. Any product substituting milk fat with cheaper vegetable oils (like palm oil) must be labelled as "Frozen Dessert". Despite the legal distinction, 70.6% of our survey respondents (see Appendix A) chose “Kwality Walls” as their preferred brand, which sells frozen dessert and uses vegetable oil.

2.6.2 The Chocolate Fat Trap

In the chocolate sector, cocoa butter is the premium fat. However, regulations permit up to 5% of vegetable fat (such as palm oil) without losing the "Chocolate" label. If this threshold is exceeded, the product must be labelled as having vegetable

fats in bold. Many popular bars in India rely on palm oil for heat stability and cost reduction.

2.6.3 The "Creme" Loophole

By using the phonetic spelling "Creme" instead of "Cream," manufacturers avoid violating dairy labelling laws while successfully misleading consumers into believing the product contains dairy-based cream when it is actually an emulsion of hydrogenated vegetable oils.

2.7 Emotional Advertising and the "Health Drink" Deception

For decades, brands like Bournvita have utilised emotional storytelling ("Taiyari Jeet Ki") to convince parents that malt-based powders are mandatory for a child's growth and immunity.

- **The Reality:** In 2023, viral analysis showed that Bournvita is roughly 50% sugar by weight. The inclusion of minor vitamins serves as a "Health Halo" to mask the high-fructose corn syrup and maltodextrin content.
- **Regulatory Action:** In April 2024, the Ministry of Commerce and Industry directed e-commerce platforms to remove Bournvita and Horlicks from the "Health Drinks" category, noting that the term has no legal definition under the FSS Act.

2.8 The Juice Mirage and the Japanese Benchmark

Indian packaged juices (e.g., Real Fruit Power) often feature high-resolution images of splashing, sliced fruits, creating a

"Health Halo" that misleads the consumer about the actual fruit content.

- **The Disparity:** Many "Real Fruit" juices contain as little as 12% fruit concentrate, with the remainder being water, sugar, and preservatives. Despite this, a high percentage of respondents preferred the juice brands that have very low real fruit content (see Appendix A).
- **The Japanese Standard:** In Japan, the *Act Against Unjustifiable Premiums and Misleading Representations* (AUPMR) enforces strict visual honesty.
    1. **100% Juice:** Only products with 100% fruit content can show a cut fruit or cross-section on the label.
    2. **5-99% Juice:** May show a whole, uncut fruit image, but any sliced representation is illegal.
    3. **<5% Juice:** No realistic fruit images are allowed; the label must use cartoon/illustrated fruit to ensure children and adults are not deceived

2.9 The CSE Honey Purity Report (2020)

In 2020, the Centre for Science and Environment (CSE) conducted a study that revealed a massive "food fraud" in the Indian honey market. Of the 13 major brands tested, including Dabur and Patanjali, 77% failed the purity tests.

- **The Deception**: Brands were using modified sugar syrups (C3 and C4) imported from China that were specifically designed to bypass FSSAI's standard laboratory tests.
- **The Mechanism**: Adulteration had "evolved" to beat traditional laboratory markers. Only advanced “Nuclear Magnetic Resonance (NMR) testing could detect the fraudulent syrups.”

- **Outcome**: While brands claimed compliance with FSSAI standards, the CSE report highlighted that FSSAI's standards were themselves outdated and unable to catch sophisticated adulteration. This case underscored the need for regulatory agencies to stay technologically ahead of the "nefarious business of adulteration."

3. The Nutritional Divide

3.1 Global Standards vs. Indian Market Compromises

One of the most insidious aspects of the modern food industry is the "Nutritional Divide"—the documented practice of multinational corporations (MNCs) utilising different ingredient profiles for identical brands in different geographical markets. Research indicates that India often receives inferior versions of products compared to those in the United States or Europe. The underlying mechanism for this divide is multifaceted. First, Indian consumers are traditionally viewed as price-sensitive, prompting brands to replace high-quality ingredients with cheaper substitutes such as palm oil, liquid glucose, and maltodextrin. Second, the "loose" regulatory environment in India, characterised by a lack of mandatory FOPL and higher permissible limits for certain additives, allows MNCs to avoid the reformulation costs they are forced to incur in the European Union or North America. Finally, companies often hide behind the defense of "catering to local tastes," arguing that Indian consumers prefer sweeter or saltier profiles to justify excessive sugar and salt levels.

A critical analysis of infant nutrition provides the clearest evidence of this double standard. A 2024 investigation by the NGO Public Eye revealed that Nestlé's infant cereal brand, Cerelac, contains significant amounts of added sugar in low- and middle-income countries (LMICs) like India and Ethiopia, while the same products in the UK, Germany, and Switzerland are sold as sugar-free. This is particularly egregious because early exposure to added sugar is a primary risk factor for lifelong obesity and diabetes.

Table 3: Comparative Analysis of Global vs. Indian Product Formulations

PRODUCT CATEGORY	INDIAN FORMULATION CHARACTERISTICS	EUROPEAN/GLOBAL FORMULATION CHARACTERISTICS	KEY "NUTRITIONAL DIVIDE" INGREDIENT
INFANT CEREAL (CERELAC)	~3.0g added sugar per serving	0.0g added sugar	Refined Sucrose / Honey
CHOCOLATE SPREADS	Higher sugar, less hazelnut, uses palm oil	Higher hazelnut %, better cocoa quality	Palm Oil / Sugar
MALTED DRINKS	High malt extract and added sugar (up to 50%)	Lower sugar, often fortified naturally	Maltodextrin / Sucrose
SOFT DRINKS (COLAS)	High-fructose corn syrup / high sucrose	Reformulated with reduced sugar in EU (4.6g/100ml)	Sugar / Artificial Sweeteners
BREAKFAST CEREALS	Twice the sodium of UK equivalents	Lower sodium due to strict regional targets	Sodium Chloride

The implications of this divide are profound. By distributing higher-sugar and higher-salt versions of products in India, MNCs are effectively exacerbating the NCD crisis in a population that is already genetically more susceptible to metabolic disorders. This "nutritional colonialism" places the burden of public health entirely on the consumer, who often lacks the literacy to decode the complex, non-standardised labels used in the Indian market.

3.2. Primary Data Analysis: Extrapolating 'Information Asymmetry'

Our survey of 51 respondents highlights a profound "Processing Paradox" (see Appendix A):

- **Trust Gap:** More than 52% of respondents feel comfortable or assume they are a better version, buying products labelled "Natural" or "Fresh," yet not even 25% felt confident enough to explain terms like “Maltodextrin”, “INS 150d”, “High Fructose Corn Syrup”, “RDA %”.
- **The Literacy Gap:** Only 47% of respondents were aware that ingredients must be listed in descending order of weight. Most assumed ingredients were listed alphabetically or randomly.
- **The Nutrition Trap:** Even among urban, educated respondents, "Sugar-Free" was interpreted as "0% Sugar" or safe for diabetics by the vast majority, with almost no knowledge of maltodextrin’s glycaemic impact.

4. Human Values & Ethics: The Moral Responsibility of Brands

4.1 Utilitarianism vs. Deontology

- **Utilitarian View:** Brands argue that using cheaper oils and sugars provides affordable calories to a massive population. However, this fails to account for the "negative

utility" of long-term healthcare costs and deaths from NCDs.

- **Deontological View:** Based on Kantian ethics, brands have an absolute duty to tell the truth. Using technical jargon to hide harmful ingredients treats the consumer as a "means to a profit" rather than an "end in themselves." The "Global Double Standard" is a direct violation of the universal right to health.

5. Conclusion and Policy Recommendations

5.1 The Case for Stricter Regulation

The evidence analysed in this report demonstrates that India's current food labelling regime is a "pro-industry" model that fails the test of public health. The "Nutritional Divide" between Indian and global markets, the complexity of INS codes, and the recurrent failures in corporate accountability all point toward a single necessity: the implementation of mandatory, interpretive Front-of-Pack Labelling (FOPL).

To mitigate the escalating public health crisis, the following strategies are proposed:

1. **Mandatory Warning Labels**: India must move away from the "Health Star Rating" (HSR) system. Experimental data clearly shows that simple, symbol-based warning labels (e.g., a red octagon for "High Sugar") are the only format that effectively alerts consumers across all literacy levels.
2. **Harmonisation of Ingredients**: FSSAI must enforce "Ingredient Equity." MNCs should be prohibited from selling versions of products in India that contain higher

levels of HFSS or harmful additives like palm oil than their versions in Europe or North America.

3. **Strict Regulation of "Health Halos"**: The use of claims like "Added Immunity" or "Real Fruit" should be legally contingent upon the product not exceeding specific thresholds for negative nutrients. A "High Sugar" product should be barred from carrying any health-related claims.
4. **Technological Upgrading of Laboratories**: As shown by the honey adulteration case, standard tests are no longer sufficient. FSSAI must mandate advanced NMR testing for high-risk categories to ensure that "Food Fraud" does not continue to bypass outdated markers.
5. **Consumer Empowerment via Digital Tools**: The FSSAI's recent moves to allow consumers to report misleading claims via an app is a positive step, but it must be backed by a transparent and time-bound grievance redressal mechanism.
6. **Plain Language Labelling (PLL):** Mandatory use of terms like "teaspoons of sugar" and functional descriptions (e.g., "Industrial Dye" instead of "INS 150").
7. **Layered Sugar Tax:** Implementing a tiered tax model where higher sugar content triggers higher taxation, incentivising reformulation.

The "fine print" of nutrition is currently a barrier to health. By transitioning to a "Warning Label" regime and closing the regulatory gaps that MNCs exploit, India can begin to reverse the tide of the NCD epidemic. The cost of inaction, measured in trillions of dollars and millions of lives, is too high to ignore. Food regulation must finally prioritise the biological reality of the consumer over the commercial reality of the manufacturer.

5.2 Conclusion: The Final Ethical Inquiry

This investigation reveals that the "Fine Print of Nutrition" is a battlefield where corporate interests often overpower public health. We have seen how "Sugar-Free" biscuits can spike insulin more than table sugar, how brown bread is often just dyed white bread, and how Indian spices face global rejections while domestic limits are relaxed. The "Nutritional Transition" in India is being fueled by a systemic asymmetry of information that leaves even the most educated consumers vulnerable to metabolic diseases.

The Ethics and Values Question:

If a food corporation possesses the scientific data to know that its product formulation contributes to a nation's metabolic decline, yet chooses to prioritise profit margins and price-sensitive ingredients over the health of its consumers, especially in regions with weaker regulatory oversight, can it ever be considered an ethical entity, or has the market fundamentally incentivised a betrayal of the social contract?

5.3 LIMITATIONS OF THE STUDY

While this report provides a detailed analysis of the Indian food sector, certain limitations must be acknowledged:

- **Sample Size:** The primary survey was conducted with 51 respondents. While insightful, this sample size is not large enough to represent the diverse socio-economic demographics of the entire Indian population.
- **Geographic Constraint:** Most respondents are from urban or semi-urban areas (Delhi-NCR); consumer behaviour in rural India regarding label literacy remains an area for further research.

- **Access to Proprietary Data:** The study relies on publicly available ingredient lists and third-party lab reports (CSE, Public Eye). Direct access to corporate manufacturing formulations was not possible.
- **Self-Reporting Bias:** Survey data is based on self-reported consumer habits, which may occasionally differ from actual real-world purchasing behaviour.

5.4 DECLARATION

We, the undersigned students of **Maharaja Surajmal Institute of Technology**, hereby declare that the project report entitled **"** *The Fine Print of Nutrition: A Deep Dive into Label Literacy, Public Health, and the Case for Stricter Food Regulation in India***"** is an authentic record of our own work carried out under the guidance of **Dr Rashmi Gupta**.

We further declare that the results embodied in this report have not been submitted to any other University or Institute for the award of any degree or diploma.
Date: April 4, 2026

Place: New Delhi

5.5 ACKNOWLEDGEMENT

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We are also thankful to **Maharaja Surajmal Institute of Technology (MSIT)** and the **GGSIP University** for providing

us with the academic environment and resources necessary to conduct this study.

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6. References

[1] FSSAI, "IIMA Front-of-Package Labelling (FoPL) Report," *Food Safety and Standards Authority of India*, Feb. 2022. [Online]. Available: https://www.fssai.gov.in

[2] A. S. Prasad et al., "Food labelling in India: a scoping review of consumer engagement, comprehension, and purchase behaviour," *Taylor & Francis / PMC*, 2024. [Online]. Available: https://pmc.ncbi.nlm.nih.gov

[3] R. Sharma and K. Gupta, "Nutrition literacy: a key factor in public health improvement in India," *Frontiers in Public Health*, 2023. [Online]. Available: https://frontiersin.org

[4] S. K. Singh, "Front-of-Package Labels on Unhealthy Packaged Foods in India: Evidence from a Randomized Field Experiment," *MDPI*, 2023. [Online]. Available: https://mdpi.com

[5] ICAR-CRIDA, "Reshaping Agriculture and Nutrition Linkages Citation page," *ICAR-Central Research Institute for Dryland Agriculture*. [Online]. Available: http://icar-crida.res.in

[6] G. Sapru, "A Star-Crossed Label: India's FOPL Delay," *The Reporters' Collective*, Oct. 05, 2025. [Online]. Available: https://www.reporters-collective.in

[7] "Regulatory Compliance Challenges for India's FSSAI," *Gates Open Research / ResearchGate*, Mar. 2024. [Online]. Available: https://www.researchgate.net

[8] A. Mukherjee et al., "Tax Policy to Promote Growth, Innovation and Investment: Carbonated Beverages Industry in India," *ICRIER*, Sep. 30, 2024. [Online]. Available: https://icrier.org

[9] "Nutrition Transition and Overnutrition in India," *NCBI PMC*, Mar. 2021. [Online]. Available: https://pmc.ncbi.nlm.nih.gov/articles/PMC8102690/

[10] The Hindu, "Nestlé baby products sold in India, Africa, Latin America have higher sugar content than in Europe, tests show," *The Hindu*, Apr. 2024. [Online]. Available: https://thehindu.com

[11] ET Edge Insights, "One in Three Food Label Claims in India Breaks the Law," *Shunyatax*, Feb. 16, 2026. [Online]. Available: https://shunyatax.in

[12] The Economic Times, "Nestle Adds Sugar to Baby Cereal in India but Not UK," *The Economic Times*, Apr. 18, 2024. [Online]. Available: https://economictimes.indiatimes.com

[13] "Hong Kong & Singapore Ban MDH and Everest over Ethylene Oxide Concerns," *Business & Human Rights Resource Centre / Wikipedia*, Apr. 2024. [Online]. Available: https://en.wikipedia.org/wiki/MDH_and_Everest_global_safety_concerns

[14] "Regulatory Implications of Banning Indian Spices: Past, Present, and Future," *SF Trainings*, Jun. 10, 2024. [Online]. Available: https://www.sftrainings.org

[15] The Hindu, "Supreme Court Asks FSSAI to Consider Warning Labels for High Sugar, Sodium, and Fat," *The Hindu /*

FoxMandal, Feb. 19, 2026. [Online]. Available: https://foxmandal.in

[16] B&B Associates LLP, "Supreme Court's Verdict on Patanjali Ayurved and Misleading Advertisements," *BNB Legal*. [Online]. Available: https://bnblegal.com

[17] AZB & Partners, "The Patanjali Journey: Supreme Court's Expanding Scrutiny on Advertising Compliance," *AZB Partners*. [Online]. Available: https://azbpartners.com

[18] Oregon State University, "Glycemic Index and Glycemic Load," *Linus Pauling Institute*. [Online]. Available: https://lpi.oregonstate.edu/mic/food-beverages/glycemic-index-glycemic-load

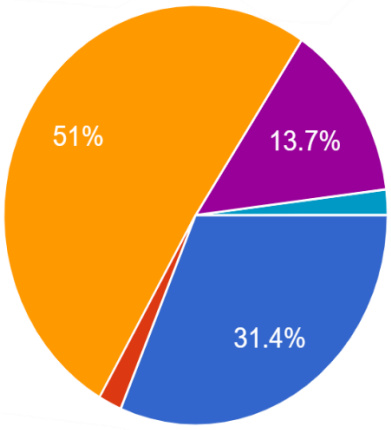
[19] The Week, "India bears the world's second-highest economic burden of diabetes; Why it matters," *The Week*, 2024. [Online]. Available: https://theweek.in

[20] Khetika, "Reading the 'Hidden Ingredients' in Everyday Indian Food Labels," *Khetika*. [Online]. Available: https://khetika.com

Appendices

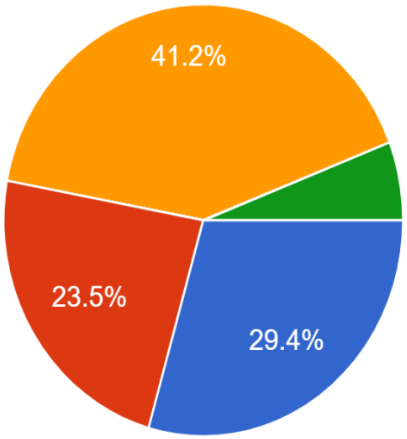
Appendix A: Raw Survey Data/Charts

1. When browsing for a packaged snack, which of these factors usually draws your attention first?



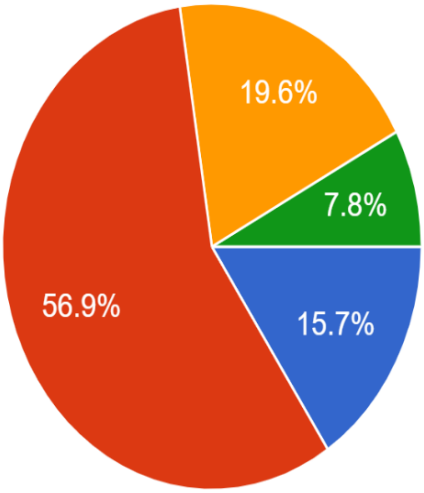
- Words on the front like "Roasted, Not Fried," "High Protein," or "No Added S…
- Attractive packaging, bright colors, or realistic food photography.
- The Brand Heritage: A name I've trusted for years (e.g., Nestle, Britannia, Amul).
- Special offers, "Extra 20% Free," or a lower price than competitors.
- I turn the pack over to see the ingredi…
- Flavours

2. If a product features a "Green Leaf" or the word "Natural" or "Fresh" on the front, what is your typical reaction?



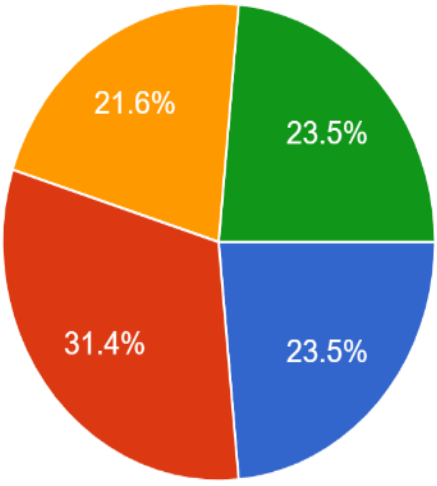
- I feel significantly more comfortable buying it for my family.
- I assume it's a better version of the product, though I don't check why.
- I feel it's likely a marketing tactic and prefer to look for the actual ingredients.
- I don't really notice these claims; I buy based on taste and flavor.

3. When you see a label claiming "0% Cholesterol" or "0% Trans Fat," how do you interpret its healthiness?



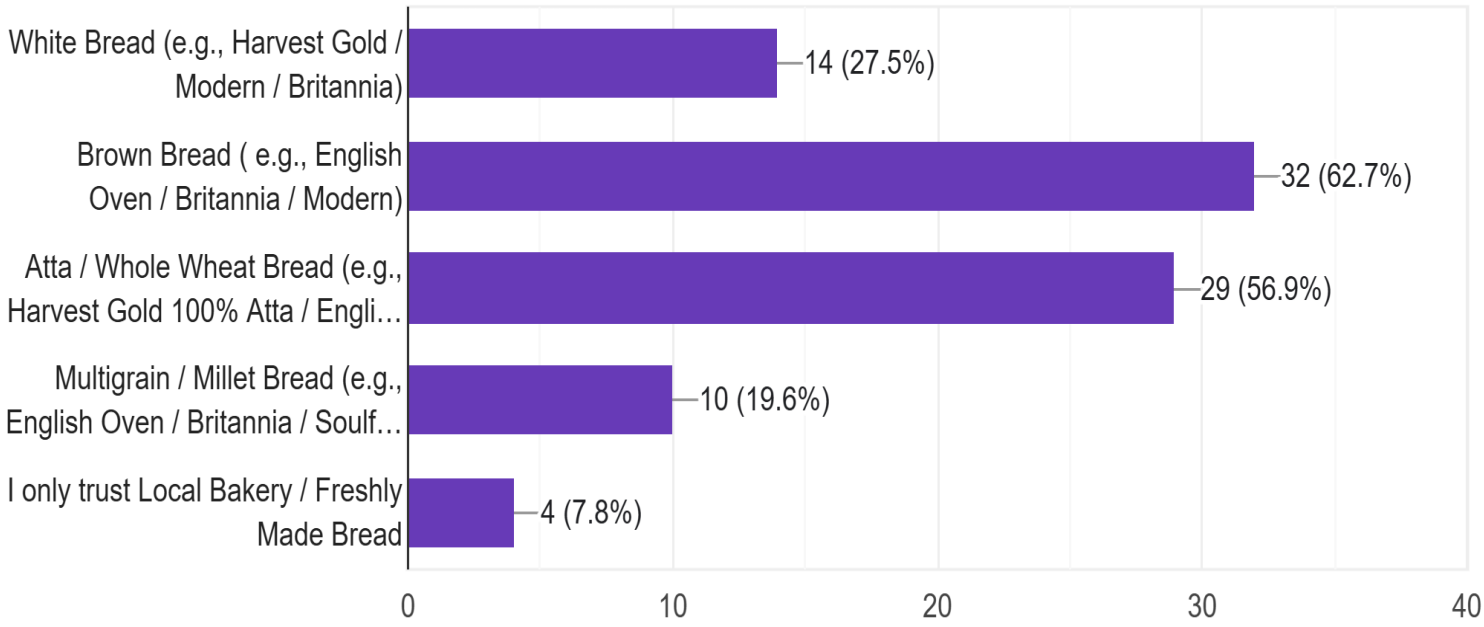
- I believe it is a heart-healthy product that I can eat regularly.
- I think it's better than other snacks
- it doesn't change my opinion.
- I don't really know what these terms mean for my long-term health.

4. When you see "Sugar-Free" or "No Added Sugar" on a packet, what is your primary understanding of its safety?

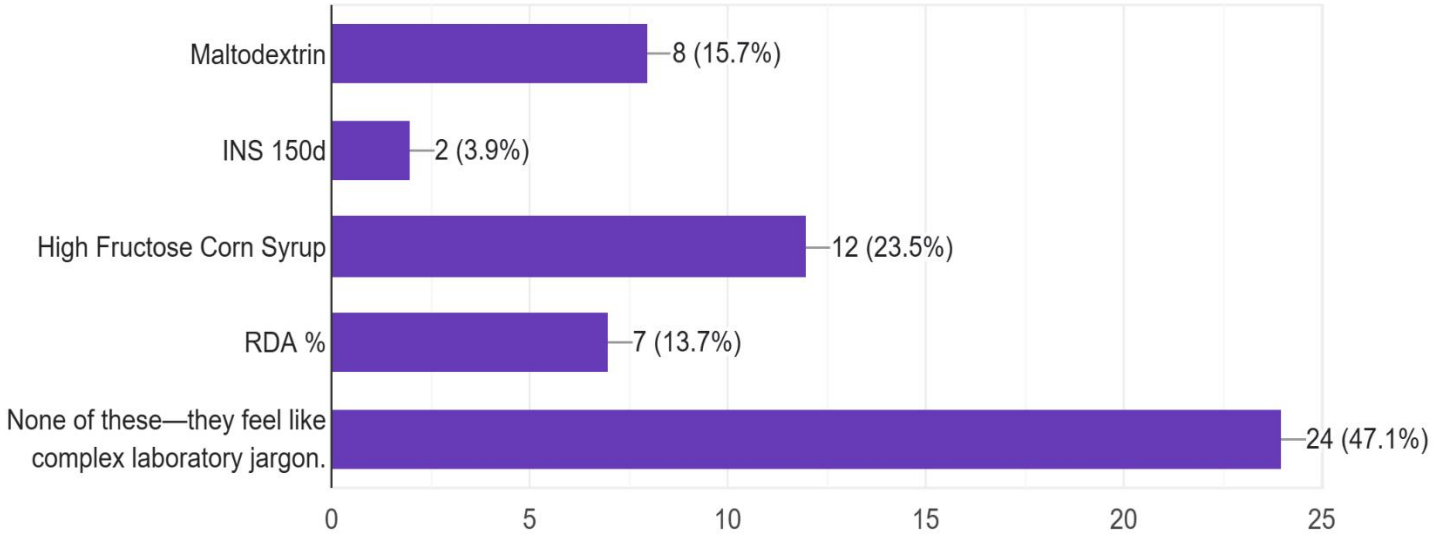


- It contains 0% sugar and is completely safe for a person with Diabetes.
- It has no "table sugar" (sucrose) but might contain "hidden" sweeteners
- It contains only natural sugars (like fruit juice concentrate) that are safer than white sugar.
- I assume it is generally healthier, but I am not familiar with the specific ingredients used.

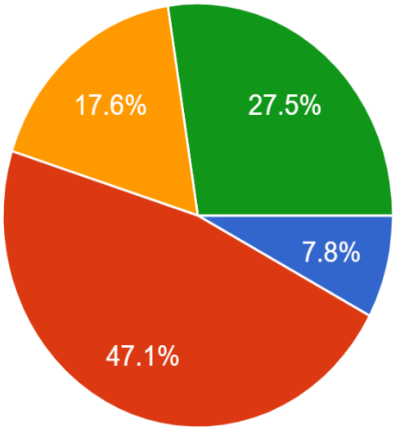
5. Which TYPE of Bread do you trust most for your family's health?



5. Which of these common industrial terms would you feel confident enough to explain to a friend? (select all that apply)

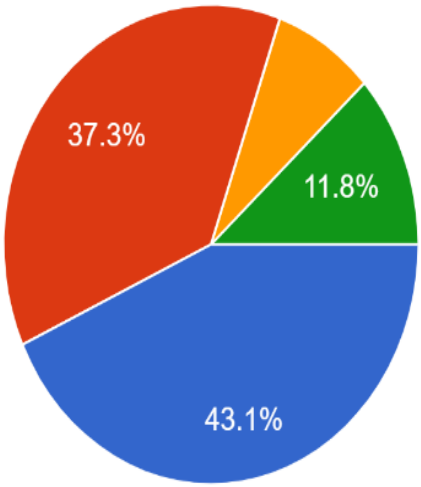


6. In what order do you believe the ingredients are listed on the back of a food packet?



- They are listed in alphabetical order for easy reading.
- They are listed from the highest quantity (by weight) to the lowest.
- They are listed randomly by the manufacturer.
- I have never noticed the order of the ingredients list.

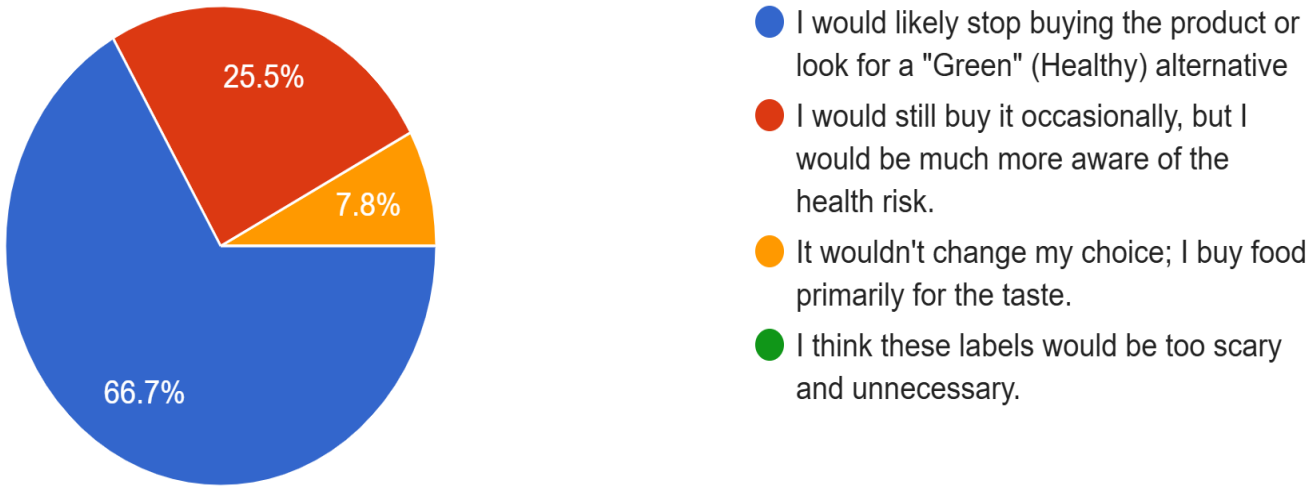
7.How would you feel if you found out your "Milk-based" product (Ice Cream/Chocolate) or "Creamy" sauce (Mayo) used Palm Oil/Vegetable Oil instead of Dairy?



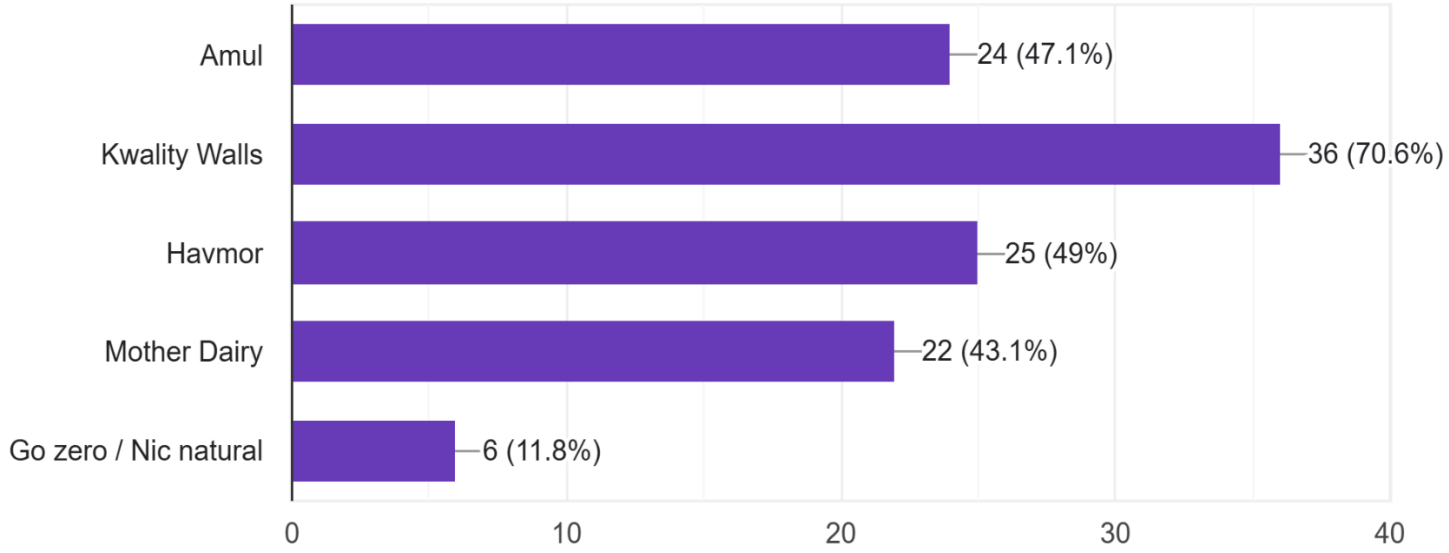
- I would be surprised; I assume these products are made primarily with milk/ cream.
- I am aware of this and specifically look for "Real Milk" or "Egg-based" labels.
- I don't check the oil source as long as the taste and texture are correct.
- I believe Vegetable Oil is actually a healthier alternative to Dairy fat

8. If the government (FSSAI) implemented a "Traffic Light" or "Warning Label" system (e.g., a Big Red Dot or a Black

Stop Sign for High Sugar) on the FRONT of the pack, how would it affect your habit?



9.Which "Ice Cream" or "Frozen Desert" brand do you Prefer?



Appendix B: Glossary of Technical Terms

TERM	DEFINITION & ETHICAL CONTEXT

INFORMATION ASYMMETRY	A state where manufacturers possess superior knowledge of a product's composition compared to the consumer, leading to biased purchasing decisions.
ULTRA-PROCESSED FOODS (UPF)	Industrial formulations of five or more ingredients (e.g., stabilisers, flavour enhancers) designed for hyper-palatability and long shelf life.
GLYCEMIC INDEX (GI)	A ranking system (0–100) indicating how quickly carbohydrates raise blood glucose. High-GI foods (>70) trigger rapid insulin spikes.
MALTODEXTRIN	A starch-derived filler with a GI of 105–110 (higher than table sugar). Often used to mask sugar content while maintaining texture.
INS CODES	The International Numbering System for food additives is a shorthand used to list chemicals (e.g., INS 150d) that many consumers find indecipherable.
NCDS	Non-Communicable Diseases, such as Type 2 diabetes and obesity, are not infectious but are driven by long-term dietary and lifestyle factors.
FOPL	Front-of-Pack Labelling: simplified nutritional icons (like "Traffic Lights" or "Warning Stars") placed on the front of a package for quick scanning.
ETHYLENE OXIDE (ETO)	A gaseous fumigant used in spices to kill bacteria. Classified as a Group 1 carcinogen, it is strictly regulated in global trade but often high in local markets.
HEALTH HALO	A marketing tactic where one "good" claim (e.g., "High in Protein") distracts the consumer from "bad" ingredients (e.g., 40% added sugar).
MAIDA VS. ATTA	Maida is refined, bleached flour stripped of fibre (High GI). Atta is whole wheat flour that retains the bran and germ, providing essential fibre.
NUTRITIONAL DIVIDE	The documented practice of MNCs using higher-quality ingredients in Western markets while selling inferior, high-sugar variants in India.
DEONTOLOGICAL ETHICS	A moral framework (Kantian) suggesting that brands have a "duty" to tell the truth, regardless of the impact on their profit margins.

Appendix C: Photographic Evidence / Screenshots



Bonn Crème Delight Biscuit Ingredient list



Kwality Walls ice cream ingredient list



English oven Brown Bread ingredient list



Real Fruit Juice Ingredient list

Appendix D: Team Photograph

